

How the Sentiment of Strategy Advice Impacts School Choice Truth-Telling in Different Information Settings

Link to Experiment: https://bates.co1.qualtrics.com/jfe/form/SV_0jn0icCXnj0jHvw

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Introduction

My final project explores the research question: *“How does the sentiment of the strategy advice influence participants’ truth-telling tendencies in school choice settings, particularly when the school assignment mechanism is unknown or partially known?”*

In this context, “truth-telling” means that applicants rank schools in a way that is consistent with their true preferences, without any consideration for strategic selection. “Sentiment” in school choice denotes if the advice actively advocate ranking schools strategically (in everyday language, “gaming the system”) or sticking to one’s true preferences. I need to conduct more research into the possible school choice mechanisms in order to exactly determine the criteria for “partially known”, but I propose a general definition of this term as only disclosing the actions taken by either the student body or the schools in response to the other party’s choices, and making sure no sophisticated participant can reason if the school choice mechanism is strategy-proof or not.

This aforementioned question is primarily based on Guillen and Hing (2014) and Koutout et al. (2021), which explore the effect of advice sentiment on school choice in a perfect-information setting. Other useful literature that supports this question include Gino and Moore (2007) which examines how participants perceive advice for tasks with varying complexity, as well as Guillen and Hakimov (2018) which presents how the type of information presented on a matching algorithm can influence participants’ truth-telling rates. Finally, Chen and Sönmez (2006) provides the necessary theoretical background and design for experimental questions on school choice.

While the literature is indeed useful for understanding the effects of advice and mechanism descriptions on participants’ truth-telling in school choice problems, no paper cited

in this project report appears to have fully answered my research question. For instance, Guillen and Hing (2014) raises a question about how the difficulty of the school allocation task may have an impact on participants' truth-telling, while Koutout et al. (2021) only tests participants' choices in response to one type of advice consistent with the optimal response to the algorithm (truth-telling for the Gale-Shapley algorithm and manipulating preferences for the Boston algorithm). Guillen and Hakimov (2018) examines the impact of the complexity and directness of information presented to participants about a matching algorithm (Top-Trading Cycles or TTC), but their experiment was not conducted in a controlled laboratory setting nor based on a school choice problem. Thus, my final project partially addresses these issues by incorporating experimental design features from these papers while making necessary modifications to factor in both the ambiguity of the algorithm and the sentiment of the advice in school choice experiments.

Apart from providing new insights on human behavior in school choice at large, this research question also contains important policy implications. Since school boards often need to consider multiple factors and accommodate multiple sources of preferences from students and their families, the mechanisms of assigning students to their schools are becoming increasingly complex. The school selection mechanisms, in many cases drawn from market design principles, may become a significant intellectual burden for the vast majority of the population without experience or specialized knowledge, particularly with people from low-income or less-educated backgrounds. As a consequence, students and their families are likely to consult advice from multiple sources when they decide on their priority rankings of schools, including sources with mixed signals like social media. By knowing the effect of strategy advice in the face of limited information, students and their families can better decide whether they should seek further

guidance in developing priority rankings of schools, while educational administrators and policy-makers can structure their communication strategy to ensure the welfare for their constituents.

Experimental Design

Participant Recruitment and Pre-Experiment Group Assignment

Before the experiment, I plan to recruit participants via CloudResearch Connect. The sample is restricted to users located in the United States and have a 95% approval rating and above. Using CloudResearch Connect's quota settings, the sample is intended to be representative of the U.S. adult population. However, in order to avoid potential biases from experimenter demand effects, participants who have taken advanced economics courses in topics related to the experiment, including game theory, experimental economics, and matching markets, are not allowed to participate in the experiment. After participants have passed a basic pre-screening questionnaire, participants will be given a link to complete the experiment online through Qualtrics.

The group assignment in the main experiment follows the “designed environment” setting in Chen and Sonmez (2006), where participants are randomly split into groups of 36. Each participant receives a random numeric ID as an integer from 1 to 36 assigned by the experimenter, which they need to enter in a software program in order to generate specific survey configurations outlined in later sections of the paper. The group composition remain consistent throughout all rounds of the experiment. Participants will not know the identity of other participants in the same group, nor the number of groups that the study contains.

The experiment has two types of between-subjects treatment: the group-level treatment regarding the school allocation algorithm and the subject-level treatment regarding the advice's sentiment. Approximately half of the groups are faced with the strategy-proof *Gale-Shapley*

(*Deferred Acceptance/DA*) algorithm and the other half received the non-strategy-proof *Boston* (*Immediate Acceptance/IA*) algorithm. My choice of algorithms is similar to the design of Koutout et al. (2021), which I will justify in a later section of this paper.

On the other hand, the advice treatment is based on Guillen and Hing (2014). Participants are randomly assigned with equal probability to one of the two advice options: the truthful advice encouraging participants to rank the schools based on their true preferences, the skeptical advice encouraging participants to manipulate their school preferences for a higher expected payoff.

Using wording from Koutout et al. (2021), each advice option is presented as follows:

Truthful advice: *Whatever ranking other participants report, you will obtain the highest possible payoff by reporting first the school for which you have the highest payoff, second the school for which you have the second highest payoff, and so on. In other words, you should rank the schools in the order of your payoffs, from high to low.*

Skeptical advice: *Be careful about the ranking you report: you will not necessarily obtain the highest possible payoff by reporting first the school for which you have the highest payoff, second the school for which you have the second highest payoff, and so on. In other words, it is not necessarily best for you to rank the schools in the order of your payoffs, from high to low.*

To avoid lying to participants and creating additional experimenter demand effects, the following sentence, adapted from Guillen and Hing (2014), is also presented before the advice:

Here is a piece of advice about school allocation procedures, which has been stylized to fit the context of this experiment. It is only here for you to consider and is not an instruction.

The School Choice Game

The main experiment lasts for three identical rounds. For each round, each group of 36 participants participates in a hypothetical admissions cycle in a fictional state.

Each participant needs to rank seven available schools A, B, C, D, E, F, and G in a specific order. Depending on the ID number, each participant is assigned to one of the school districts (also listed as A-G in accordance with the specific school that it hosts), along with the specific payoff value upon admission to each of the seven schools, which may assist them in determining their school preference rankings.

Participants indicate their order of preference for each of the seven schools by dragging and dropping the names of each specific school (A through G) to their desired position, where a higher placement of a school in the list indicates a stronger preference for that school. A sample experiment interface, which contains the full instructions for the baseline school choice game, is shown in Figure 1 of Appendix 1.

After a participant has submitted their initial school rankings for the round, a piece of advice is shown on the participant's screen. The participant is then prompted to decide whether they want to change their school preference rankings given the advice, and if so, they need to submit a new set of rankings. A sample experiment interface for this step is shown in Figure 2 of Appendix 1.

The Information Treatment

For each participant, the amount of information about the school allocation algorithm that the participant's group is subjected to serves as the within-subjects treatment for the experiment.

At the start of Round 1, participants are only given basic information about the school choice game itself and how they are expected to submit their school ranking preferences.

Participants receive no information of what algorithm their final school allocation is based upon, nor the identity of other members in their group.

Before declaring their school preference rankings in Round 2, participants are informed that each school in the school choice game has its own priority order to determine whether certain applicants are admitted in an admission cycle, and the exact mechanism of the priority order itself. The wording to describe priority orders for both school choice algorithms mentioned in the experiment, Boston and Gale-Shapley, is taken nearly verbatim from Chen and Sonmez (2006), with appropriate adjustments as I adapt the instructions from an originally paper-based experiment to an online Qualtrics survey.

In Round 3, participants are fully informed of the mechanism of the school allocation algorithm that their group is subjected to, although they are not informed of the algorithm's exact name (Boston/Immediate Acceptance or Gale-Shapley/Deferred Acceptance). Similar to the information treatment for Round 2, the wording to describe both algorithms is also taken nearly verbatim from Chen and Sonmez (2006).

Attention and Comprehension Check Measures

In order to generate better quality data as participants fully read and comprehend the instructions, as well as reducing possibilities of bots or AI-generated answers in the survey experiment, participants are required to complete an attention or comprehension check process before they enter the school choice game for each round.

For Round 1, as there is no information about the algorithm disclosed to participants, each participant only need to take a basic attention check question before proceeding to the main school choice game. Participants are removed from the survey and are not allowed to receive any

compensation for participating in the experiment if they answer the attention check question incorrectly.

For Rounds 2 and 3, participants need to answer a comprehension check question based on the content of disclosed information before each round. In Round 2, participants answer a basic multiple-choice question related to the criteria that each school use to determine the priority order of each participant. A correct answer within the first attempt gives the participant an extra compensation of \$1.

If the participant does not answer the question correctly within the first attempt, the software program will display the correct answer and an explanation of why this particular answer is correct. After that, the participant has another chance to answer the question correctly for an extra \$0.5. However, if the participant does not pass the comprehension check within two attempts, they will be removed from the survey and are not allowed to receive any compensation for participating in the experiment.

In Round 3, the participant need to apply the algorithm that they have learned from the instructions to answer a hypothetical school choice problem that allocates five students to three school slots. The wording of the school choice problem is identical to that of the example presented in the full information treatment. For each correct allocation, the participant receives an extra \$0.25 in compensation. If the participant allocates all students to their correct schools within one single attempt, they will also earn a bonus payment of \$1. The experiment interface of the first attempt for this comprehension check question is shown in Figure 3 of Appendix 1.

If one of the participants' allocations is incorrect, the software program will display the set of all five correct allocations and a detailed explanation of the allocation process used in the question. After that, the participant has another chance to solve the school allocation problem,

this time receiving an extra \$0.1 in compensation for each student allocation that they submitted incorrectly in the first attempt but correctly in the second attempt. If the participant allocates all students to their correct schools in the second attempt (given the answers and explanations), they will also earn a bonus payment of \$0.5. Due to the complexity allocated with school choice allocation procedures, a participant will not be removed from the survey if they does not answer the allocation problem fully correctly within two attempts.

Additional Measures

After each participant has completed three rounds of the school choice game, the participant is redirected to another page to complete a brief problem-solving assessment which questions are taken verbatim from Guillen and Hing (2014). Participants need to solve three mathematical reasoning problems, each having a time limit of 30 seconds. A timer indicating the number of seconds left to answer the question will also be displayed on the participant's screen. For each correct answer, the participant earns an additional \$1. The experiment interface for a sample question in the problem-solving assessment is shown in Figure 4 of Appendix 1.

Finally, participants will be redirected to an optional demographic and post-experiment feedback survey, which answers have no effect on participants' final compensation for the experiment. The demographic questions cover age, gender identity, race/ethnicity, income, level of education, field of study, prior experience with economics, and familiarity with matching market problems. The post-experiment feedback questions, partially adapted from Guillen and Hing (2014) and Koutout et al. (2021), ask whether the participant has read and followed the advice, their perception of the advice and the purpose of the experiment, and whether the participant thinks that there is a particular strategy to rank their schools to get the best possible compensation.

Post-Experiment Instructions

After all participants in a group have completed a survey, all group members' specific school rankings (after the advice treatment) declared in one randomly chosen round are entered into a solver coded in a programming language such as Python to determine the final school allocation and corresponding payoff for each participant in that group, as suggested by Chen and Sonmez (2006).

In addition to payoffs from the school choice game, each participant also receives a base payment of \$3 for completing the survey, along with additional payment for answering attention check, comprehension check, and problem-solving questions correctly throughout the experiment. The total amount of compensation and instructions to receive the compensation will be announced via email to each participant within two weeks after all participants in the group have completed the survey.

Design Justification

First, my within-subjects design of gradually releasing details about the school choice mechanisms carries external validity, because it simulates the experience and thought processes that students and their families often undergo before submitting their school preferences. More specifically, as non-specialized people take time to learn about the allocation algorithm and consult advice at different periods of learning, they have a better understanding about the chances of getting into highly preferred schools with different strategies of stating their preferences.

It can also be argued that this experiment contains order effects impacting participants' decision-making, but as stated before, the rounds' progression from neither information nor advice to full information plus advice reflects reality more accurately, acting as a soft nudge for

participants to reflect on their choices in the previous round and make necessary adjustments to maximize their payoff. Moreover, it is extremely difficult and perhaps unethical to encourage participants to forget what they have learned about the allocation mechanism once the information has already been presented.

Second, the Boston and Gale-Shapley algorithms in my experimental design fundamentally differ in their strategy-proofness, but are quite similar in terms of popularity (historically and currently implemented in multiple school districts) as well as levels of complexity (the algorithm's full description requires similar amounts of information).¹ Therefore, my choice to compare these two allocation mechanisms serves as between-subjects treatments to account for necessary algorithm-specific factors that may influence participants' truth-telling tendencies. Furthermore, the comparison between Boston and Gale-Shapley algorithms also contributes to existing literature (Koutout et al., 2021) by incorporating the effects of algorithm details and incorrect advice on participants' decision-making.

Finally, I mentioned the variation in the design of attention and comprehension check questions throughout each stage of the experiment. This aspect does not only reflects the increase in the minimum level of information about the school allocation mechanisms that participants need to process by each round, but also allows the researcher to control for the diverse range in participants' logical reasoning and problem-solving abilities, which determines whether they rely on conscious calculations or intuition to assign certain rankings for each school. For instance, "sophisticated" participants can be defined as people who successfully solve the hypothetical

¹ I am assuming that the majority of strategy-proof algorithms (such as Gale-Shapley or TTC) are more complex than non-strategy-proof algorithms, in part because designing a strategy-proof algorithm involves addressing multiple potential loopholes that can be strategically exploited by sophisticated players.

school allocation problem for Round 3's comprehension check within the first attempt, and answer at least two problem-solving questions correctly within the given time frame.

Analysis Plan

Hypotheses

My experiment contains five hypotheses numbered from H1 to H5. While H1, H2, and H3 are designed to verify the existing literature on the subject, H4 and H5 answer my own question about the impact of the strategy advice's sentiment on participants' truth-telling tendencies in school choice problems with varying levels of information. These include:

H1: If no information are supplied, participants are equally likely to submit their true preferences. However, in the partial and full information treatment, participants are more likely to tell the truth when they are presented with the Gale-Shapley algorithm than in the Boston algorithm (Chen and Sonmez, 2006).

H2: The advice has a significant impact on swaying the truth-telling tendencies of participants towards its intended sentiment (Guillen and Hing, 2014; Koutout et al., 2021). That is, participants are more likely to submit their true preferences after being exposed to an advice encouraging participants to be truthful, and less likely to submit their true preferences after being exposed to an advice encouraging otherwise.

H3: In terms of magnitude, the skeptical advice has a more significant impact on swaying the truth-telling tendencies of participants towards its intended sentiment than the truthful advice (Guillen and Hing, 2014). That is, the increase in the truth-telling rate for participants who are exposed to an advice encouraging them to be truthful is lower than the decrease in the truth-telling rate for participants who are exposed to an advice encouraging otherwise.

H4: The effect of the advice on swaying the truth-telling tendencies of participants towards its intended sentiment decreases in magnitude as participants are exposed to more information about their allocation algorithms, regardless of which algorithm they are assigned to.

H5: Given the inherent complexity of the allocation algorithms (Guillen and Hing, 2014), the advice's sentiment has a higher impact on participants' truth-telling tendencies than the information presented on the allocation algorithm.

Outcome Variables

As stated in the previous section, the experimental design from Chen and Sonmez (2006) has already dictated each participant's "true preferences" through its payoff table and its random assignment of hypothetical profiles to participants. The only gap not addressed in evaluating the truth-telling tendencies of participants in this design is whether participants' submitted rankings are consistent with their true preferences.

The main outcome variable is the proportion of participants whose submitted school rankings are truthful to their own preferences. According to Koutout et al. (2021), if a participant is assigned to the Boston algorithm, a ranking is actually truthful when all seven schools are ranked in descending order of their payoff, listing the school with the highest payoff as the first preference to the school with the lowest payoff as the last preference.

On the other hand, if the participant is assigned to the Gale-Shapley algorithm, a ranking is actually truthful if the participant rank all out-of-district schools with a payoff higher than the participant's district school in descending order of their payoff, then ranks their district school after (Koutout et al., 2021). The discrepancy happens because the unique design of the Gale-Shapley algorithm ensures that participants are guaranteed a school assignment at least as good

as their own district school, regardless of their final preference ranking of all seven schools in question (Chen and Sonmez, 2006).

Sample Size

The experiment is performed on groups of 36 participants each, with each group being assigned to one of the two school allocation algorithms, and each participant receiving one of the two advice treatments. To ensure that all four combinations of school allocation algorithm and advice treatment is available in the study, I plan to recruit the baseline sample of at least $36 \cdot 4 = 144$ participants.

Due to the inherent complexity of the school allocation algorithms, I also expect that the experiment is likely to have a high exclusion rate as participants struggle to stay focused on the survey. Considering that CloudResearch Connect allows a maximum rejection rate of 20% (CloudResearch, 2025), I also plan to recruit an additional $(144/80\%) - 144 = 36$ participants for the contingency sample. These participants are not allowed to take the survey immediately after they have passed the pre-screening questionnaire, but they will be prompted to take the survey if certain participants drop out of the experiment. In this case, the participants will enter the survey with a specific, non-random configuration of algorithm and advice to ensure that the allocation process runs seamlessly with a full group of 36 participant IDs.

Future research may use specific power analysis to determine the desired sample size in a more statistically rigorous way.

Proposed Statistical Analyses

As a general note, for each statistical test outlined in this section, including two-sample t-tests for differences in means and two-sample z-tests for differences in proportions, my analyses will include the means or proportions obtained from each subsample, the associated standard

error and the 95% confidence interval, and the p-value of the test. In case where a regression model is needed, I plan to include each explanatory variable's coefficient, its standard error and 95% confidence interval, along with whether the coefficient for that particular variable is significantly different from zero.

All proposed statistical analyses in this paper are conducted with a significance threshold of $\alpha = 0.05$. The p-values of all statistical test results and regression coefficients are either reported fully to at least two decimal places, or reported according to one of the three levels commonly used in asterisk reporting systems for Stata-generated results: $*p < 0.1$ (slightly significant), $**p < 0.05$ (moderately significant), and $***p < 0.01$ (extremely significant).

Baseline Analyses

To test for H1, I will conduct two-proportion z-tests for each of the three possible levels of information that participants know about the school allocation algorithm used in determining their group members' payoffs. For the no information treatment, the test is two-tailed to determine if the truth-telling rate of participants whose payments are allocated based on the Boston algorithm is significantly different from that of participants subjected to the Gale-Shapley algorithm. On the other hand, the test for the partial and full information treatments is one-tailed, testing whether the truth-telling rate of participants under the Gale-Shapley algorithm is higher than that of participants under the Boston algorithm. This result is expected because of inherent differences in strategy-proofness of the two algorithms in question.

My analysis and statistical reporting plan for H2 is similar to that for H1, as I also aim to use two-proportion z-tests to compare truth-telling rates between two samples of participant responses for each type of advice sentiment. However, the tests for H2 are designed to compare truth-telling rates before and after the advice treatment is shown to participants. The tests are

one-tailed, with the direction depending on the sentiment of the advice itself. For the truthful treatment, I expect the truth-telling rate after the advice is greater than that before the advice. In contrast, for the skeptical treatment, I expect the truth-telling rate before the advice is greater than that after the advice is shown.

Swaying Effects of the Advice

For each subsample of participants assigned to a specific advice treatment, I use two different formulas to generate an overall indicator of the effect of the advice on swaying participants' truth-telling tendencies to its intended direction:

For the subsample receiving truthful advice:

$$\text{Swaying effect} = (\text{Truth-telling rate } \textit{after} \text{ advice}) - (\text{Truth-telling rate } \textit{before} \text{ advice})$$

For the subsample receiving skeptical advice:

$$\text{Swaying effect} = (\text{Truth-telling rate } \textit{before} \text{ advice}) - (\text{Truth-telling rate } \textit{after} \text{ advice})$$

The difference in how the swaying effects are calculated between two subsamples is a result of comparing the impact of two contradictory advice treatments. A positive swaying effect indicates that the advice has successfully swayed participants towards its intended direction (more truth-telling after the truthful advice, and less truth-telling after the skeptical advice).

To answer H3, I will use a two-sample t-test to compare the swaying effects in the subsample of participants who receive the truthful advice and those receiving the skeptical advice. The test is one-tailed, with the expectation that the swaying effect in the skeptical treatment is significantly higher than that in the truthful treatment.

H4 is actually an extension of H3 with an additional within-subjects information treatment variable. We first compute the swaying effect of the advice in subsamples of participant responses for each combination of advice sentiment and level of information

supplied. Then, for each round of the experiment (each information treatment), we take the average of the swaying effects between participants receiving the truthful advice and those receiving the skeptical advice. This average is weighted by the number of participants in each advice treatment subsample to avoid an unusually large subsample influencing the overall swaying effect of the advice regardless of its sentiment.

Since there are three different information treatments available (no, partial, and full information), we conduct three pairwise t-tests between two of the information treatments. In order for H4 to be verified, the advice's swaying effect should be greatest in the no information treatment, second greatest in the partial information treatment, and lowest in the full information treatment.

Comparing the Effect of Advice Sentiment and Information on Truth-Telling

Finally, to answer H5, I propose an individual fixed effects regression model, which can be set up as follows:

$$\begin{aligned} tell_truth_{it} = & \beta_0 + \beta_1 truth_advice_i + \beta_2 info_level_t \\ & + \beta_3 truth_advice_i * partial_info_t + \beta_4 algorithm_i + \gamma_i + \epsilon_{it} \end{aligned}$$

The outcome variable is a binary variable that answers whether the participant is submitting their true preferences. The key explanatory variables in this model are the advice treatment dummy (*truthful_advice*, whether the advice is truthful or not) and the information treatment (*info_level*, an ordinal variable coded as 0 if no information, 1 if partial information, and 2 if full information).

I have also included the interaction term between the truthful advice treatment and whether the information is presented. This dummy variable equals 1 if the advice is both truthful

and the participant is exposed to at least partial information about the school allocation algorithm that determines their payoffs.

The interaction term's coefficient β_3 being significantly different from zero indicates that the advice's sentiment and the level of information presented have different effects on determining participants' truth-telling tendencies, which is the necessary but not sufficient condition for H5. To fully support H5, the coefficient for the effect of the advice's sentiment (β_1) should also be both significantly different from zero and greater than the coefficient for the effect of the information treatment (β_2).

As an additional robustness check, I also include additional control variables in my model. One important control variable is the allocation algorithm dummy to control for algorithm's inherent strategy-proofness (*algorithm*, coded as 0 for Boston, 1 for Gale-Shapley). The individual fixed effects actually appear in the term γ_i to control for participant-specific, time-invariant characteristics: whether the participant is “sophisticated” (they can answer difficult comprehension check and problem-solving questions), and demographic variables such as age, gender identity, income, level of education, and number of Economics courses taken at undergraduate level.²

² Further robustness checks can be conducted for hypotheses from H1 to H4 by converting the corresponding statistical tests into OLS regression models similar to that of H5. The outcome variable is always the probability that the participant submits their true preferences, and additional participant fixed effects are included along with the key explanatory variables.

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Appendix 1: The Experiment Interface

For demonstration, these figures show the experiment interface for a participant who is assigned an ID number of 1, the truthful advice treatment, and the Boston school allocation algorithm. In the actual experiment, the exact school district, payoff configuration, and survey instructions vary from participant to participant based on the specific group and between-subjects treatment to which the participant is assigned.

In most figures, the experiment interface is presented as a long single page that cannot be fully displayed on one computer screen. Participants read the complete instructions and learn about the possible choices by scrolling to the bottom of the page.

When multiple images appear in a figure, they represent different views of the page that together cover the entire instructions. The individual images should be read from left to right and from top to bottom.

Figure 1

Main Experiment Interface of the Baseline School Choice Game Before the Advice Treatment

Recall: You are participant #1 and you live within the school district of School A. Recall: Your payoff amount depends on the school slot you hold at the end of the experiment. Payoff amounts are outlined below: – You will be paid \$13 if you hold a slot at school A at the end of the experiment. – You will be paid \$16 if you hold a slot at school B at the end of the experiment. – You will be paid \$9 if you hold a slot at school C at the end of the experiment. – You will be paid \$2 if you hold a slot at school D at the end of the experiment. – You will be paid \$5 if you hold a slot at school E at the end of the experiment. – You will be paid \$11 if you hold a slot at school F at the end of the experiment. – You will be paid \$7 if you hold a slot at school G at the end of the experiment.	Please rank your preferences for each of the schools (A through G) by dragging and dropping each school's name to the desired position. A higher placement of a school in the list indicates a stronger preference for that school. <i>The Next (→) button will appear 60 seconds after this page is opened. Please take your time and make the best possible decision.</i> <table style="width: 100%; border-collapse: collapse;"> <tr><td style="width: 30px; text-align: center;">1</td><td>A</td></tr> <tr><td style="text-align: center;">2</td><td>B</td></tr> <tr><td style="text-align: center;">3</td><td>C</td></tr> <tr><td style="text-align: center;">4</td><td>D</td></tr> <tr><td style="text-align: center;">5</td><td>E</td></tr> <tr><td style="text-align: center;">6</td><td>F</td></tr> <tr><td style="text-align: center;">7</td><td>G</td></tr> </table>	1	A	2	B	3	C	4	D	5	E	6	F	7	G
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Figure 2*Main Experiment Interface of the Baseline School Choice Game with the Advice Treatment*

Here is a piece of advice about school allocation procedures, which has been stylized to fit the context of this experiment. It is only here for you to consider and is not an instruction: <i>"Whatever ranking other participants report, you will obtain the highest possible payoff by reporting first the school for which you have the highest payoff, second the school for which you have the second highest payoff, and so on. In other words, you should rank the schools in the order of your payoffs, from high to low."</i> Your current preference ranking for each school, based on your response to the previous question, is indicated in the table below: <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">School</th> <th style="text-align: left;">Preference ranking</th> </tr> </thead> <tbody> <tr><td>A</td><td>2</td></tr> <tr><td>B</td><td>1</td></tr> <tr><td>C</td><td>4</td></tr> <tr><td>D</td><td>7</td></tr> <tr><td>E</td><td>6</td></tr> <tr><td>F</td><td>3</td></tr> <tr><td>G</td><td>5</td></tr> </tbody> </table>	School	Preference ranking	A	2	B	1	C	4	D	7	E	6	F	3	G	5	Recall: You are participant #1 and you live within the school district of School A. Recall: Your payoff amount depends on the school slot you hold at the end of the experiment. Payoff amounts are outlined below: – You will be paid \$13 if you hold a slot at school A at the end of the experiment. – You will be paid \$16 if you hold a slot at school B at the end of the experiment. – You will be paid \$9 if you hold a slot at school C at the end of the experiment. – You will be paid \$2 if you hold a slot at school D at the end of the experiment. – You will be paid \$5 if you hold a slot at school E at the end of the experiment. – You will be paid \$11 if you hold a slot at school F at the end of the experiment. – You will be paid \$7 if you hold a slot at school G at the end of the experiment.
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F	3																
G	5																

You will have another chance to change your preference rankings for schools A through G given the advice by dragging and dropping each school's name to the desired position. Remember, a higher placement of a school in the list indicates a stronger preference for that school. If you do not want to change any of your preference rankings, select the option "I do not want to change my preferences". <i>The Next (→) button will appear 40 seconds after this page is opened. Please take your time and make the best possible decision.</i>	☐ I do not want to change my preferences. 1 A 2 B 3 C 4 D 5 E 6 F 7 G
--	---

Note: When the participant selects the option "I do not want to change my preferences", the drag-and-drop interface for submitting the new preference rankings will disappear from the page. To make the ranking interface appear again, the participant can simply uncheck that option.

Figure 3*Experiment Interface for the Comprehension Check Question Before Round 3, First Attempt*

In order to check your understanding of the school allocation method that you have learned, we ask you to allocate students to schools in the following hypothetical scenario.

There are five students with ID numbers 1, 2, 3, 4, and 5, and three schools named X, Y, and Z. There are two slots available each at School X and School Z, and one slot available at School Y. Residents of districts are indicated in the table below.

School	Slot 1	Slot 2	District Residents
X	☐	☐	1 2
Y	☐		3
Z	☐	☐	4 5

If a school places multiple students at the same priority level, ties will be resolved using the following lottery order: **1 → 2 → 3 → 4 → 5**.

The students submit the following school rankings:

	1 st Choice	2 nd Choice	3 rd (Last) Choice
Student 1	X	Y	Z
Student 2	Z	X	Y
Student 3	Y	Z	X
Student 4	Z	Y	X
Student 5	Y	X	Z

Please determine the corresponding school allocation for each student using the allocation method that you have learned. Each correct allocation will earn you a bonus payment of **\$0.25**. Additionally, you will also earn a bonus **\$1** if you allocate all five students to their correct schools at this point in the experiment.

Feel free to solve the problem at your own pace. You are highly recommended to use scratch paper while answering the question.

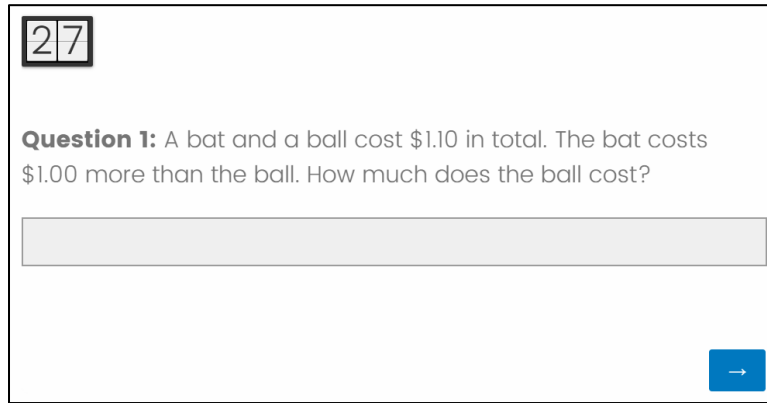
The Next (→) button will appear 60 seconds after this page is opened. Please take your time, read the question carefully, and enter your best guess.

	School X	School Y	School Z
Student 1	☐	☐	☐
Student 2	☐	☐	☐
Student 3	☐	☐	☐
Student 4	☐	☐	☐
Student 5	☐	☐	☐

Note: The participant answers the question by clicking the corresponding buttons to select exactly one of three possible school allocations for each of the five students.

Figure 4

Main Experiment Interface for a Problem-Solving Assessment Question



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Question 1: A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost?

→

Note: A countdown clock is displayed on the participant's screen, which shows the number of seconds left for the participant to answer the question and counts down at each second. Since Qualtrics generally does not allow automatic response submission, the participant needs to click on the Next (→) button to have their answer recorded.